

MINI PROJECT REPORT

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Project Title : **"Menu & Modifier Sales Analysis and Customer Behavior Insights Using Power BI"**

Project Domain : Hospitality & Food Service Analytics

Tools Used : Microsoft Excel, Power BI Desktop, DAX

Submission Date : 21-02-2026

Mentor Name : KUMARAN M

Raw Dataset Link : https://drive.google.com/drive/folders/1bO4jm21LEpc918sErN61zqdV89wnOh8Y?usp=drive_link

Cleaned Dataset Link : https://docs.google.com/spreadsheets/d/1Q5xdAO2jJhbFquTN3kPKqCjD-Q4_wvox/edit?usp=drive_link&oid=111235392724903046158&rtpof=true&sd=true

Executive Summary

This project focuses on analyzing hotel sales and customer behavior using Power BI. The dataset was cleaned in Excel and transformed for analysis. Key performance indicators such as Total Revenue, Orders, Average Order Value, and Modifier Usage Percentage were calculated using DAX.

The dashboard provides insights into store performance, discount impact, service mode trends, and customer customization patterns. The analysis helps identify revenue drivers and opportunities for menu optimization.

Problem Statement

The hotel business generates sales from various menu items, service modes, and add-on modifiers. However, without proper analysis, it becomes difficult to identify revenue drivers, customer behavior patterns, and the effectiveness of modifiers in increasing sales.

The objective of this project is to analyze sales data to identify trends, evaluate modifier usage impact, understand peak performance periods, and provide insights that can help improve overall revenue performance and operational decision-making.

Dataset Overview

The dataset used for this project contains transactional-level hotel sales data. Each row represents an individual menu item sold within a customer order.

The original dataset consisted of **1,744 rows** and **14 columns**. After performing data cleaning and transformation in Microsoft Excel, the dataset was refined and expanded to **17 columns** to support enhanced analysis and reporting.

Original Dataset Preview

A	B	C	D	E	F	G	H	I	J	K	L	M	N
order_id	store_id	transaction_datetime	business_day	daypart	service_mode	menu_item	modifier	quantity	unit_price	discount	tax	total_amount	payment_type
120000	102	05-07-25 6:14	05-07-25	Breakfast	Drive-Thru	Bottled Water		1	1.48	0.15	0.1	1.43	Cash
120001	201	05-07-25 11:48	05-07-25	Lunch	Dine-In	Chicken Sandwich		1	5.34	0	0.45	5.79	Card
120002	403	13-04-25 8:05	13-04-25	Breakfast	Takeout	Bottled Water	Light Ice	1	1.49	0.15	0.1	1.44	Cash
120003	301	25-06-25 6:42	25-06-25	Breakfast	Drive-Thru	Egg Muffin	No Cheese	1	3.62	0	0.34	3.96	Cash
120004	301	01-08-25 12:58	01-08-25	Lunch	Drive-Thru	Spicy Chicken Combo		2	9.17	0	1.38	19.72	Card
120005	501	12-07-25 12:32	12-07-25	Lunch	Dine-In	Double Cheeseburger		1	5.64	0	0.37	6.01	Card
120006	101	26-07-25 17:56	26-07-25	Afternoon	Delivery	Onion Rings		1	3.59	0.54	0.29	3.34	MobilePay
120007	403	21-07-25 16:48	21-07-25	Afternoon	Takeout	Soft Drink (Medium)		1	1.8	0.27	0.11	1.64	Cash
120008	102	02-06-25 23:08	02-06-25	Late Night	Delivery	Chicken Sandwich Extra Sauce		2	5.22	0	0.79	11.73	Card
120009	402	12-07-25 5:21	12-07-25	Breakfast	Drive-Thru	Hash Browns		1	2	0	0.12	2.12	Card
120010	103	19-04-25 21:15	19-04-25	Dinner	Takeout	Double Cheeseburger		1	5.55	0	0.33	5.88	Card
120011	102	16-07-25 11:08	16-07-25	Morning	Dine-In	Apple Pie		1	1.53	0	0.1	1.63	Card
120012	201	08-06-25 8:13	08-06-25	Breakfast	Takeout	Sausage Biscuit	Sausage Patty	2	3.1	0	0.64	9.24	Card
120013	103	13-07-25 10:37	13-07-25	Morning	Takeout	Coffee	Oat Milk	2	1.51	0	0.27	4.69	Cash
120014	201	11-04-25 12:37	11-04-25	Lunch	Dine-In	Spicy Chicken Combo		1	8.43	0	0.67	9.1	Card
120015	102	22-05-25 9:06	22-05-25	Breakfast	Delivery	Egg Muffin		1	3.1	0	0.23	3.33	Card
120016	403	27-04-25 9:19	27-04-25	Breakfast	Drive-Thru	Egg Muffin		1	2.99	0	0.22	3.21	MobilePay
120017	403	16-04-25 11:05	16-04-25	Lunch	Takeout	Cheeseburger		1	4.59	0		5	Card
120018	403	23-06-25 9:35	23-06-25	Breakfast	Takeout	Sausage Biscuit	Extra Egg	2	2.81	0	0.55	8.17	Card
120019	103	10-07-25 19:51	10-07-25	Dinner	Dine-In	Chicken Sandwich Extra Sauce		1	4.87	0	0.37	5.49	Card
120020	201	14-04-25 11:22	14-04-25	Morning	Drive-Thru	Coffee	Sugar-Free Syrup	1	2.18	0	0.19	2.67	MobilePay
120021	101	07-07-25 6:37	07-07-25	Breakfast	Dine-In	Sausage Biscuit	Cinnamon	1	2.98	0	0.26	3.34	Cash
120022	403	11-04-25 17:47	11-04-25	Afternoon	Dine-In	Ice Cream Cone	Cinnamon	1	1.65	0	0.13	1.88	Card
120023	201	11-04-25 14:43	11-04-25	Lunch	Drive-Thru	Spicy Chicken Sandwich	No Pickles	1	5.09	0	0.41	5.5	Cash
120024	103	04-07-25 17:42	04-07-25	Afternoon	Takeout	Ice Cream Cone		1	1.58	0	0.11	1.69	Card
120025	401	28-05-25 19:45	28-05-25	Dinner	Takeout	Chicken Sandwich Extra Cheese		1	5.43	0	0.56	6.49	Card

Original dataset before cleaning (14 columns, 1,744 rows).

The dataset includes the following key columns:

Column Name	Description
Order_ID	Unique identifier for each customer order
Transaction_Date	Date on which the transaction occurred
Business_Day	Operational reporting date of the business
Store_ID	Identifier for store location
Service_Mode	Mode of service (Dine-in, Takeaway, etc.)
Menu_Item	Name of the menu item sold
Quantity	Number of items ordered
Unit_Price	Price per individual item
Discount	Discount amount applied
Tax	Tax amount applied
Total_Amount	Final revenue amount after discount and tax
Modifier	Additional customization selected by customer
Payment_Type	Mode of payment used
Day_Of_Week	Derived column for weekday analysis
Time	Extracted time for time-based analysis
Gross_Amount	Revenue amount before financial adjustments

Additionally, the dataset contains both **Transaction_Date** and **Business_Day** columns. In this dataset, both fields contained identical date values and therefore did not require separate treatment during analysis.

The cleaned dataset was then imported into Power BI, where DAX measures were created to calculate key performance indicators and analytical metrics for dashboard development.

Data Cleaning & Transformation

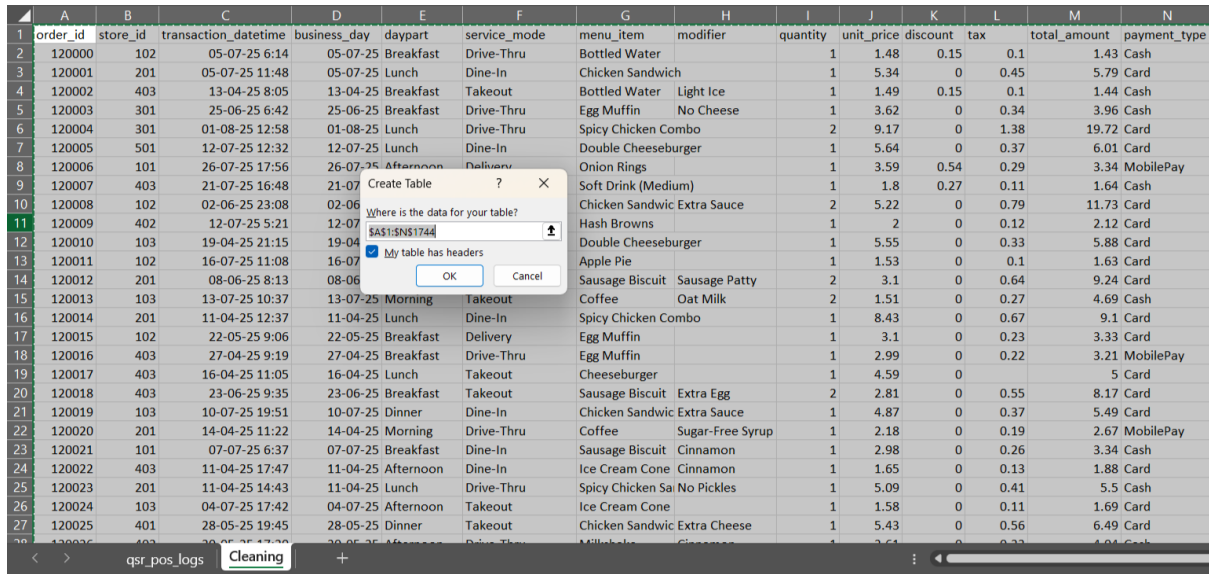
Before performing any modifications, the original dataset was preserved by creating a duplicate worksheet named “**Cleaning**” in Microsoft Excel. All data cleaning and transformation activities were carried out in this new sheet to maintain the integrity of the raw data.

The original dataset contained **14 columns**, which were later refined and expanded to **17 columns** after cleaning and transformation.

The following data preparation steps were performed:

1. Conversion to Structured Table Format

The dataset in the “Cleaning” sheet was converted into a structured Excel table format to improve data organization, enable efficient formula usage, and maintain consistency.

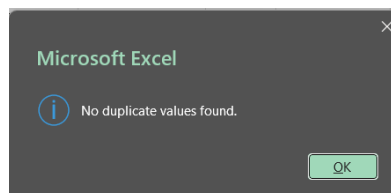
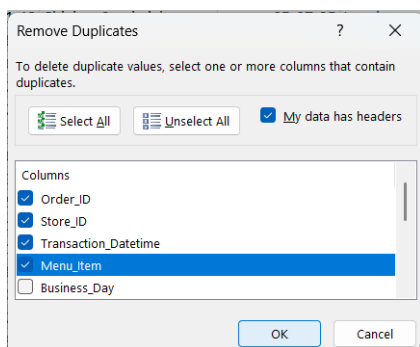


	A	B	C	D	E	F	G	H	I	J	K	L	M	N
	order_id	store_id	transaction_datetime	business_day	daypart	service_mode	menu_item	modifier	quantity	unit_price	discount	tax	total_amount	payment_type
2	120000	102	05-07-25 6:14	05-07-25	Breakfast	Drive-Thru	Bottled Water		1	1.48	0.15	0.1	1.43	Cash
3	120001	201	05-07-25 11:48	05-07-25	Lunch	Dine-In	Chicken Sandwich		1	5.34	0	0.45	5.79	Card
4	120002	403	13-04-25 8:05	13-04-25	Breakfast	Takeout	Bottled Water	Light Ice	1	1.49	0.15	0.1	1.44	Cash
5	120003	301	25-06-25 6:42	25-06-25	Breakfast	Drive-Thru	Egg Muffin	No Cheese	1	3.62	0	0.34	3.96	Cash
6	120004	301	01-08-25 12:58	01-08-25	Lunch	Drive-Thru	Spicy Chicken Combo		2	9.17	0	1.38	19.72	Card
7	120005	501	12-07-25 12:32	12-07-25	Lunch	Dine-In	Double Cheeseburger		1	5.64	0	0.37	6.01	Card
8	120006	101	26-07-25 17:56	26-07-25	Afternoon	Delivery	Onion Rings		1	3.59	0.54	0.29	3.34	MobilePay
9	120007	403	21-07-25 16:48	21-07-25	Afternoon	Delivery	Soft Drink (Medium)		1	1.8	0.27	0.11	1.64	Cash
10	120008	102	02-06-25 23:08	02-06-25	Evening	Delivery	Chicken Sandwich Extra Sauce		2	5.22	0	0.79	11.73	Card
11	120009	402	12-07-25 5:21	12-07-25	Morning	Delivery	Hash Browns		1	2	0	0.12	2.12	Card
12	120010	103	19-04-25 21:15	19-04-25	Evening	Delivery	Double Cheeseburger		1	5.55	0	0.33	5.88	Card
13	120011	102	16-07-25 11:08	16-07-25	Morning	Delivery	Apple Pie		1	1.53	0	0.1	1.63	Card
14	120012	201	08-06-25 8:13	08-06-25	Morning	Delivery	Sausage Biscuit	Sausage Patty	2	3.1	0	0.64	9.24	Card
15	120013	103	13-07-25 10:37	13-07-25	Morning	Takeout	Coffee	Oat Milk	2	1.51	0	0.27	4.69	Cash
16	120014	201	11-04-25 12:37	11-04-25	Lunch	Dine-In	Spicy Chicken Combo		1	8.43	0	0.67	9.1	Card
17	120015	102	22-05-25 9:06	22-05-25	Breakfast	Delivery	Egg Muffin		1	3.1	0	0.23	3.33	Card
18	120016	403	27-04-25 9:19	27-04-25	Breakfast	Drive-Thru	Egg Muffin		1	2.99	0	0.22	3.21	MobilePay
19	120017	403	16-04-25 11:05	16-04-25	Lunch	Takeout	Cheeseburger		1	4.59	0		5	Card
20	120018	403	23-06-25 9:35	23-06-25	Breakfast	Takeout	Sausage Biscuit	Extra Egg	2	2.81	0	0.55	8.17	Card
21	120019	103	10-07-25 19:51	10-07-25	Dinner	Dine-In	Chicken Sandwich Extra Sauce		1	4.87	0	0.37	5.49	Card
22	120020	201	14-04-25 11:22	14-04-25	Morning	Drive-Thru	Coffee	Sugar-Free Syrup	1	2.18	0	0.19	2.67	MobilePay
23	120021	101	07-07-25 6:37	07-07-25	Breakfast	Dine-In	Sausage Biscuit	Cinnamon	1	2.98	0	0.26	3.34	Cash
24	120022	403	11-04-25 17:47	11-04-25	Afternoon	Dine-In	Ice Cream Cone	Cinnamon	1	1.65	0	0.13	1.88	Card
25	120023	201	11-04-25 14:43	11-04-25	Lunch	Drive-Thru	Spicy Chicken Sai No Pickles		1	5.09	0	0.41	5.5	Cash
26	120024	103	04-07-25 17:42	04-07-25	Afternoon	Takeout	Ice Cream Cone		1	1.58	0	0.11	1.69	Card
27	120025	401	28-05-25 19:45	28-05-25	Dinner	Takeout	Chicken Sandwich Extra Cheese		1	5.43	0	0.56	6.49	Card
28	120026	103	20-06-25 17:40	20-06-25	Evening	Drive-Thru	Milkshake	Chocolate	1	3.61	0	0.33	4.04	Cash

2. Duplicate Validation

To ensure data integrity, a duplicate check was performed using the **Remove Duplicates** feature in Excel based on the following columns:

- Order_ID
- Store_ID
- Transaction_Date
- Menu_Item



No duplicate records were found.

3. Text Standardization

Inconsistent text formatting was identified in certain columns. Using the **Find and Replace** feature along with text case correction, values were standardized into proper case format to maintain uniformity across the dataset.

Find	Replace
Find what: order_id	Replace with: Order_ID

Find	Replace
Find what: store_id	Replace with: Store_ID

Find	Replace
Find what: business_day	Replace with: Business_Day

Find	Replace
Find what: service_mode	Replace with: Service_Mode

Find	Replace
Find what: modifier	Replace with: Modifier

Find	Replace
Find what: unit_price	Replace with: Unit_Price

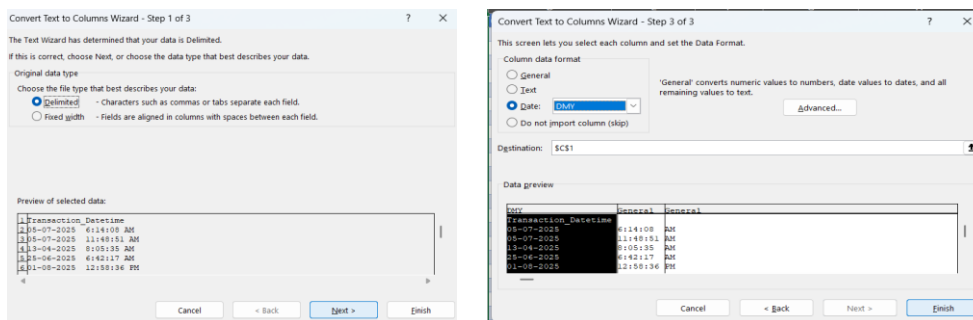
Find	Replace
Find what: tax	Replace with: Tax

4. Data Type Formatting

After text standardization, appropriate data formats were assigned to all columns (Date, Time, Currency, and Numeric formats) to ensure accuracy and consistency for analysis.

5. Date and Time Separation

The **Transaction_Time** column originally contained both date and time values. Using the **Text to Columns** feature in Excel, the time component (including AM/PM format) was separated into a new column named **Time** for improved time-based analysis.



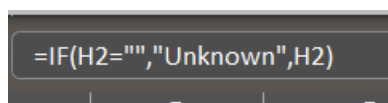
Additionally, a new column **Day_Of_Week** was derived from the **Business_Day** column to support weekday trend analysis.

```
=TEXT([@[Business_Day]],"dddd")
```

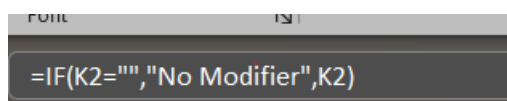
6. Handling Missing Values with Logical Assumptions

Blank values were addressed carefully based on contextual understanding:

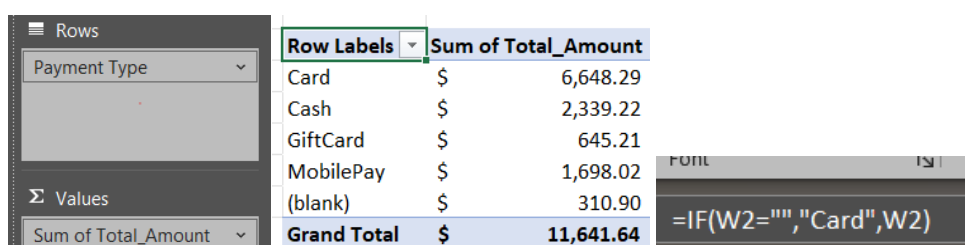
- **Service_Mode:** Blank values were replaced with “Unknown”, as the service type could not be inferred from other fields.



- **Modifier:** Blank values were replaced with “No Modifier”, since modifier selection is independent of other attributes.



- **Payment_Type:** Missing values were analyzed using a Pivot Table to identify the most frequent payment method and filled accordingly.

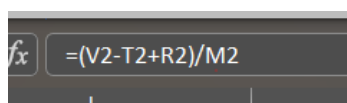


Row Labels	Sum of Total_Amount
Card	\$ 6,648.29
Cash	\$ 2,339.22
GiftCard	\$ 645.21
MobilePay	\$ 1,698.02
(blank)	\$ 310.90
Grand Total	\$ 11,641.64

7. Unit Price Recalculation

Although there were no blank values in the Unit_Price column, some values were inaccurate. The correct unit price was recalculated using the formula:

$$\text{Unit_Price} = (\text{Total_Amount} - \text{Tax} + \text{Discount}) / \text{Quantity}$$

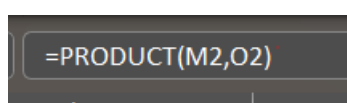


This ensured financial accuracy and consistency.

8. Gross Amount Calculation

A new column **Gross_Amount** was created using:

$$\text{Gross_Amount} = \text{Quantity} \times \text{Unit_Price}$$



This represents the amount before tax and discount adjustments.

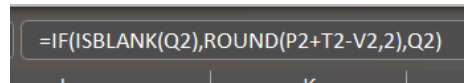
9. Conditional Corrections for Financial Columns

To maintain internal consistency, conditional formulas were applied:

- **Discount**

If blank:

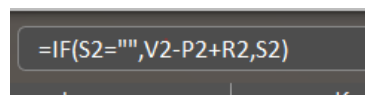
=IF(ISBLANK(Discount), ROUND(Gross_Amount + Tax – Total_Amount, 2), Discount)



- **Tax**

If blank:

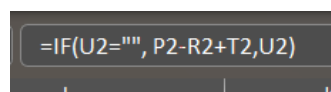
=IF(Tax = "", Total_Amount – Gross_Amount + Discount, Tax)



- **Total_Amount**

If blank:

=IF(Total_Amount = "", Gross_Amount – Discount + Tax, Total_Amount)



These corrections ensured that all financial values aligned mathematically.

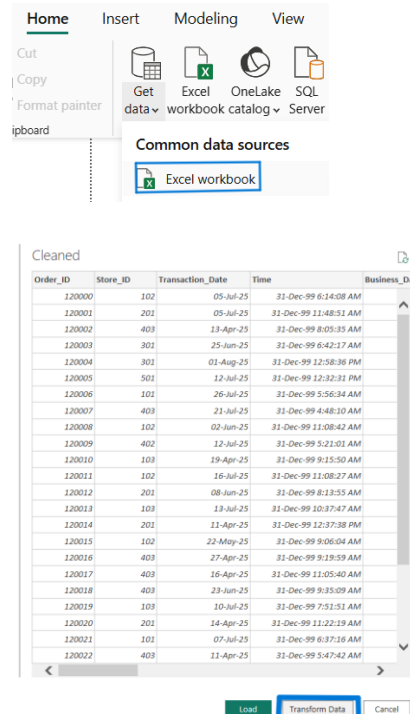
Final Cleaned Dataset

Order ID	Order No.	Transaction Date	Time	Business Day	Day of Week	Order Type	Service Type	Item Name	Modifier	Payment Type	Quantity	Unit Price	Gross Amount	Discount	Tax	Total Amount
120000	102	05-07-25	6:14 AM	05-07-25 Saturday	Breakfast	Drive-Thru	Bottled Water	No Modifier	Cash		1	1.48	1.48	0.15	0.1	1.43
120001	201	05-07-25	11:48 AM	05-07-25 Saturday	Lunch	Dine-in	Chicken Sandwich	No Modifier	Card		1	5.34	5.34	0	0.45	5.79
120002	403	13-04-25	8:05 AM	13-04-25 Sunday	Breakfast	Takeout	Bottled Water	Light Ice	Cash		1	1.49	1.49	0.15	0.1	1.44
120003	301	25-06-25	6:42 AM	25-06-25 Wednesday	Breakfast	Drive-Thru	Egg Muffin	No Cheese	Cash		1	3.62	3.62	0	0.34	3.96
120004	301	01-08-25	12:58 PM	01-08-25 Friday	Lunch	Drive-Thru	Spicy Chicken Combo	No Modifier	Card		2	9.17	18.34	0	1.38	19.72
120005	501	12-07-25	12:32 PM	12-07-25 Saturday	Lunch	Dine-in	Double Cheeseburger	No Modifier	Card		1	5.64	5.64	0	0.37	6.01
120006	101	26-07-25	5:56 AM	26-07-25 Saturday	Afternoon	Delivery	Onion Rings	No Modifier	MobilePay		1	3.59	3.59	0.54	0.29	3.94
120007	403	21-07-25	4:48 AM	21-07-25 Monday	Afternoon	Takeout	Soft Drink (Medium)	No Modifier	Cash		1	1.80	1.8	0.27	0.11	1.64
120008	102	02-06-25	11:08 AM	02-06-25 Monday	Late Night	Delivery	Chicken Sandwich	Extra Sauce	Card		2	5.47	10.94	0	0.79	11.73
120009	402	12-07-25	5:21 AM	12-07-25 Saturday	Breakfast	Drive-Thru	Hesh Browns	No Modifier	Card		1	2.00	2	0	0.12	2.12
120010	103	19-04-25	9:15 AM	19-04-25 Saturday	Dinner	Takeout	Double Cheeseburger	No Modifier	Card		1	5.55	5.55	0	0.33	5.88
120011	102	16-07-25	11:08 AM	16-07-25 Wednesday	Morning	Dine-in	Apple Pie	No Modifier	Card		1	1.53	1.53	0	0.1	1.63
120012	201	08-06-25	8:13 AM	08-06-25 Sunday	Breakfast	Takeout	Sausage Biscuit	Sausage Patty	Card		2	4.30	8.6	0	0.64	9.24
120013	103	15-07-25	10:37 AM	15-07-25 Sunday	Morning	Takeout	Coffee	Oat Milk	Cash		2	2.21	4.42	0	0.27	4.69
120014	201	11-04-25	12:37 PM	11-04-25 Friday	Lunch	Dine-in	Spicy Chicken Combo	No Modifier	Card		1	8.43	8.43	0	0.67	9.10
120015	102	22-05-25	9:06 AM	22-05-25 Thursday	Breakfast	Delivery	Egg Muffin	No Modifier	Card		1	3.10	3.1	0	0.23	3.33
120016	403	27-04-25	9:19 AM	27-04-25 Sunday	Breakfast	Drive-Thru	Egg Muffin	No Modifier	MobilePay		1	2.99	2.99	0	0.22	3.21
120017	403	16-04-25	11:05 AM	16-04-25 Wednesday	Lunch	Takeout	Cheeseburger	No Modifier	Card		1	1.48	1.48	0	0.1	1.58
120018	403	23-06-25	9:35 AM	23-06-25 Monday	Breakfast	Takeout	Sausage Biscuit	Extra Egg	Card		2	3.81	7.62	0	0.55	8.17
120019	103	10-07-25	7:51 AM	10-07-25 Thursday	Dinner	Dine-in	Chicken Sandwich	Extra Sauce	Card		1	5.12	5.12	0	0.37	5.49
120020	201	14-04-25	11:22 AM	14-04-25 Monday	Morning	Drive-Thru	Coffee	Sugar-free Syrup	MobilePay		1	2.48	2.48	0	0.19	2.67
120021	101	07-07-25	6:37 AM	07-07-25 Monday	Breakfast	Dine-in	Sausage Biscuit	Cinnamon	Cash		1	3.08	3.08	0	0.26	3.34
120022	403	11-04-25	5:47 AM	11-04-25 Friday	Afternoon	Dine-in	Ice Cream Cone	Cinnamon	Card		1	1.75	1.75	0	0.13	1.88
120023	201	11-04-25	2:43 AM	11-04-25 Friday	Lunch	Drive-Thru	Spicy Chicken Sandwich	No Pickles	Cash		1	5.09	5.09	0	0.41	5.50
120024	103	04-07-25	5:42 AM	04-07-25 Friday	Afternoon	Takeout	Ice Cream Cone	No Modifier	Card		1	1.58	1.58	0	0.11	1.69
120025	401	28-05-25	7:45 AM	28-05-25 Wednesday	Dinner	Takeout	Chicken Sandwich	Extra Cheese	Card		1	5.93	5.93	0	0.56	6.49
120026	402	30-05-25	5:20 AM	30-05-25 Friday	Afternoon	Drive-Thru	Milksake	Cinnamon	Cash		1	3.71	3.71	0	0.33	4.04
120027	402	05-07-25	4:04 AM	05-07-25 Saturday	Afternoon	Dine-in	French Fries	Small Size	Card		1	2.45	2.45	0	0.17	2.62
120028	201	21-06-25	6:54 AM	21-06-25 Saturday	Dinner	Drive-Thru	Cheeseburger	No Modifier	Card		1	4.29	4.29	0	0.36	4.65
120029	202	06-08-25	2:21 AM	06-08-25 Wednesday	Lunch	Delivery	Chicken Sandwich	No Onions	Card		3	5.28	15.84	0	1.55	17.19
120030	103	22-06-25	7:13 AM	22-06-25 Sunday	Breakfast	Takeout	Sausage Biscuit	No Modifier	Cash		1	2.99	2.99	0.15	0.27	3.11
120031	101	09-07-25	7:53 AM	09-07-25 Wednesday	Dinner	Delivery	Cheeseburger Combo	No Modifier	Card		2	9.42	18.84	0	1.41	20.25
120032	103	27-07-25	10:59 AM	27-07-25 Sunday	Late Night	Drive-Thru	Spicy Chicken Sandwich	Extra Cheese	MobilePay		1	6.63	6.63	0	0.48	7.11
120033	501	26-06-25	9:42 AM	26-06-25 Thursday	Dinner	Dine-in	Cheeseburger	No Modifier	Cash		1	4.14	4.14	0	0.33	4.47
120034	202	14-04-25	1:49 AM	14-04-25 Monday	Afternoon	Takeout	Ice Tea	No Modifier	MobilePay		1	2.00	2	0	0.15	2.15
120035	101	29-06-25	12:31 PM	29-06-25 Sunday	Lunch	Takeout	Chicken Sandwich	No Modifier	Card		1	5.94	5.94	0	0.43	6.37
120036	402	07-08-25	11:18 AM	07-08-25 Thursday	Lunch	Takeout	Spicy Chicken Combo	No Modifier	Card		1	9.39	9.39	0	0.89	10.28
120037	201	06-05-25	12:12 PM	06-05-25 Tuesday	Lunch	Drive-Thru	Chicken Sandwich Combo	Spicy	Card		1	8.01	8.01	0	0.58	8.59
120038	101	31-09-25	12:44 PM	31-09-25 Saturday	Late Night	Drive-Thru	Cheeseburger	No Modifier	Cash		1	4.05	4.05	0	0.32	4.37
120039	401	28-05-25	11:09 AM	28-05-25 Wednesday	Lunch	Drive-Thru	Cheeseburger	No Modifier	MobilePay		1	4.56	4.56	0	0.33	4.89

Final cleaned dataset after all transformations (17 columns).

Data Import & Power Query Processing

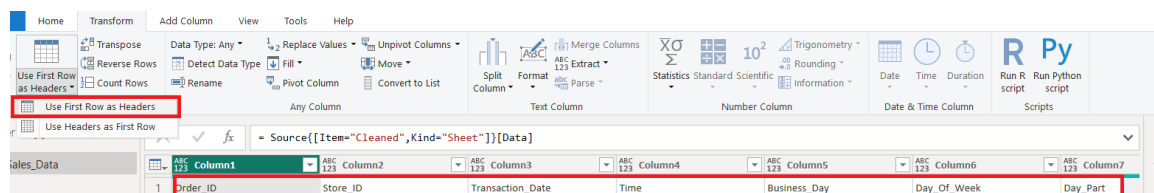
After completing data cleaning in Microsoft Excel, the finalized dataset was imported into Power BI using the Get Data → Excel Workbook option.



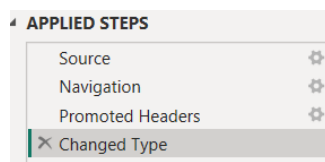
Before loading the data into the model, basic validation was performed in Power Query Editor to ensure structural consistency.

The following steps were carried out in Power Query:

- Promoted the first row as column headers.



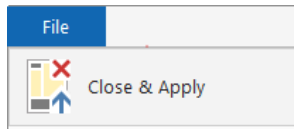
- Verified and corrected data types for relevant columns (Date, Time, Numeric, and Currency).



- Performed a final validation check to confirm data accuracy before loading.

Since the dataset was already cleaned and structured in Excel, no additional transformations were required in Power Query.

The dataset was then loaded into Power BI for DAX measure creation and dashboard development.



DAX Measures & Calculations

After importing the cleaned dataset into Power BI, DAX measures were created to calculate key performance indicators (KPIs) and perform advanced business analysis.

The following measures were developed:

1. Total Revenue

Purpose: Calculates the overall revenue generated from all transactions.

Logic: SUM of Total_Amount column.

```
1 Total_revenue = SUM(Sales_Data[Total_Amount])
```

2. Total Quantity

Purpose: Calculates the total number of items sold.

Logic: SUM of Quantity column.

```
Total_Quantity = SUM(Sales_Data[Quantity])
```

3. Total Orders

Purpose: Calculates the total number of unique customer orders.

Logic: COUNT of Order_ID.

```
1 Total_Orders = COUNT(Sales_Data[Order_ID])
```

4. Unique Menu Items

Purpose: Counts the number of distinct menu items sold.

Logic: DISTINCTCOUNT of Menu_Item.

```
Unique_Menu_Items = DISTINCTCOUNT(Sales_Data[Menu_Item])
```

5. Unique Modifiers

Purpose: Counts the number of distinct modifiers used.

Logic: DISTINCTCOUNT of Modifier.

```
Unique_Modifier = DISTINCTCOUNT(Sales_Data[Modifier])
```

6. Average Order Value (AOV)

Purpose: Calculates the average revenue generated per order.

Logic: Total Revenue ÷ Total Orders.

```
AOV = DIVIDE([Total_revenue],[Total_Orders])
```

7. Orders With Modifier

Purpose: Counts total orders where at least one modifier was selected.

Logic: DISTINCTCOUNT of Order_ID where Modifier ≠ "No Modifier".

```
Orders_With_Modifier = CALCULATE([Total_Orders],Sales_Data[Modifier]<> "No Modifier")
```

8. Orders Without Modifier

Purpose: Counts total orders where no modifier was selected.

Logic: DISTINCTCOUNT of Order_ID where Modifier = "No Modifier".

```
Orders_Without_Modifier = CALCULATE([Total_Orders],Sales_Data[Modifier] = "No Modifier")
```

9. Modifier Usage %

Purpose: Shows the percentage of orders that include modifiers.

Logic:

$(\text{Orders With Modifier} \div \text{Total Orders}) \times 100$

```
Modifier_Usage % = DIVIDE([Orders_With_Modifier],[Total_Orders],0)
```

10. Revenue With Modifier

Purpose: Calculates total revenue generated from orders that include modifiers.

Logic: SUM of Total_Amount where Modifier $\neq$ "No Modifier".

```
Revenue_With_Modifier = CALCULATE([Total_revenue],Sales_Data[Modifier] <> "No Modifier")
```

11. AOV With Modifier

Purpose: Calculates the average order value for orders that include modifiers.

Logic: Revenue With Modifier $\div$ Orders With Modifier.

```
Revenue_Without_Modifier = CALCULATE([Total_revenue],Sales_Data[Modifier] = "No Modifier")
```

12. AOV Without Modifier

Purpose: Calculates the average order value for orders without modifiers.

Logic: Revenue Without Modifier $\div$ Orders Without Modifier.

```
AOV_With_Modifier = DIVIDE([Revenue_With_Modifier],[Orders_With_Modifier],0)
```

13. AOV Difference

Purpose: Identifies the difference in spending behavior between customers who used modifiers and those who did not.

Logic:

AOV Without Modifier – AOV With Modifier.

```
AOV_Difference = [AOV_Without_Modifier] - [AOV_With_Modifier]
```

This measure helps evaluate whether modifiers increase or decrease average customer spending.

Dashboard Development Overview

After completing data cleaning in Excel and creating calculated measures using DAX in Power BI, an interactive dashboard was developed to visualize business performance and customer behavior effectively.

The dashboard was designed with a storytelling approach and divided into two pages:

- **Page 1 – Sales Overview**

- **Page 2 – Customer Behavior & Trends Analysis**

The main goal of the dashboard is to transform raw transactional data into meaningful business insights through interactive visualizations.

Page 1 – Sales Overview

Page Objective

The objective of this page is to provide a high-level summary of overall sales performance across stores, service modes, and time periods.

This page helps management quickly understand key performance indicators and identify sales trends.

KPI Cards Used & Purpose

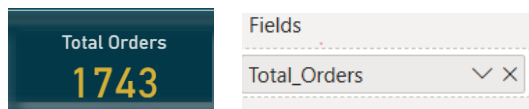
1. Total Revenue

- **Purpose:** Displays the total sales revenue generated for the selected time period and filters.
- **Business Value:** Helps management monitor overall business performance instantly.



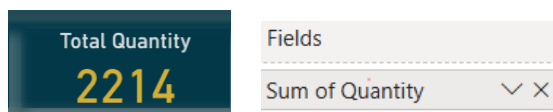
2. Total Orders

- **Purpose:** Shows the total number of orders placed.
- **Business Value:** Measures customer purchase activity and demand.



3. Total Quantity Sold

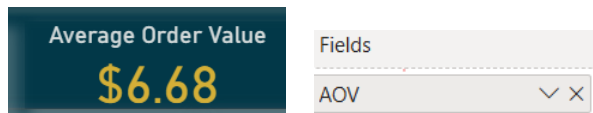
- **Purpose:** Displays total number of items sold.
- **Business Value:** Helps understand product movement and sales volume.



4. Average Order Value (AOV)

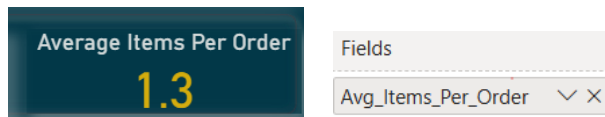
- **Purpose:** Calculates the average revenue per order.

- **Business Value:** Indicates customer spending behavior.



5. Average Items per Order

- **Purpose:** Shows the average number of items purchased in a single order.
- **Business Value:** Helps analyze customer buying patterns and upselling performance.



Slicers Used & Purpose

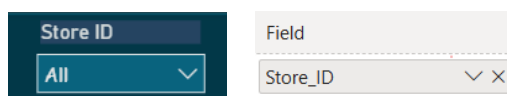
1.Date Slicer

- Allows filtering of data by specific date range.
- Helps analyze performance over different time periods.



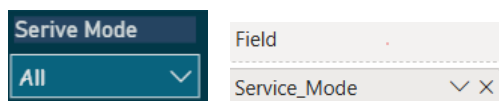
2.Store ID Slicer

- Filters sales data by store.
- Helps compare performance across multiple stores.



3.Service Mode Slicer

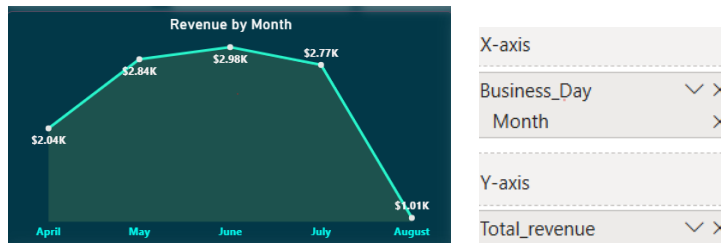
- Filters data by service type (Dine-in, Takeaway, Delivery).
- Helps understand revenue contribution by operational channel.



Visual Analysis

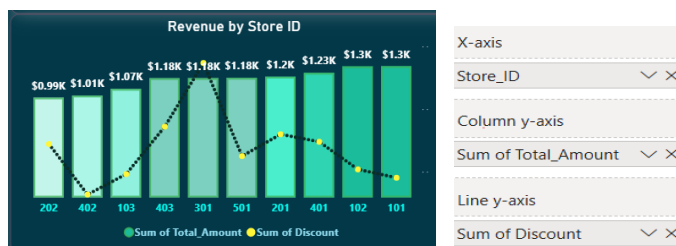
1. Revenue by Month (Line Chart)

- Shows monthly sales trend.
- Helps identify seasonal patterns and growth trends.



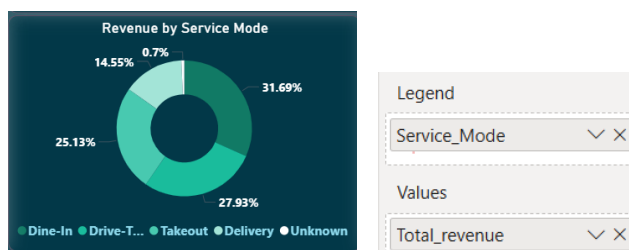
2. Revenue by Store (Bar Chart)

- Compares performance of different stores.
- Identifies top and low-performing locations.



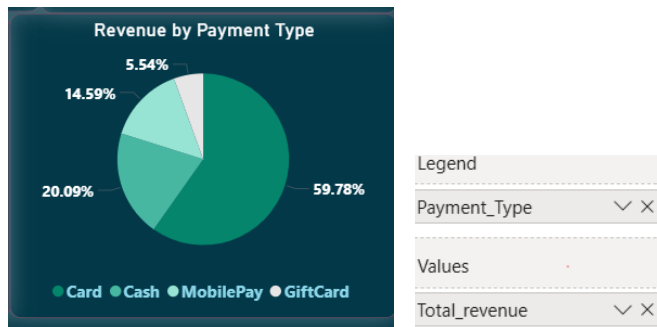
3. Revenue by Service Mode (Bar/Column Chart)

- Shows contribution of each service mode.
- Helps evaluate operational efficiency.



4. Revenue by Payment Type

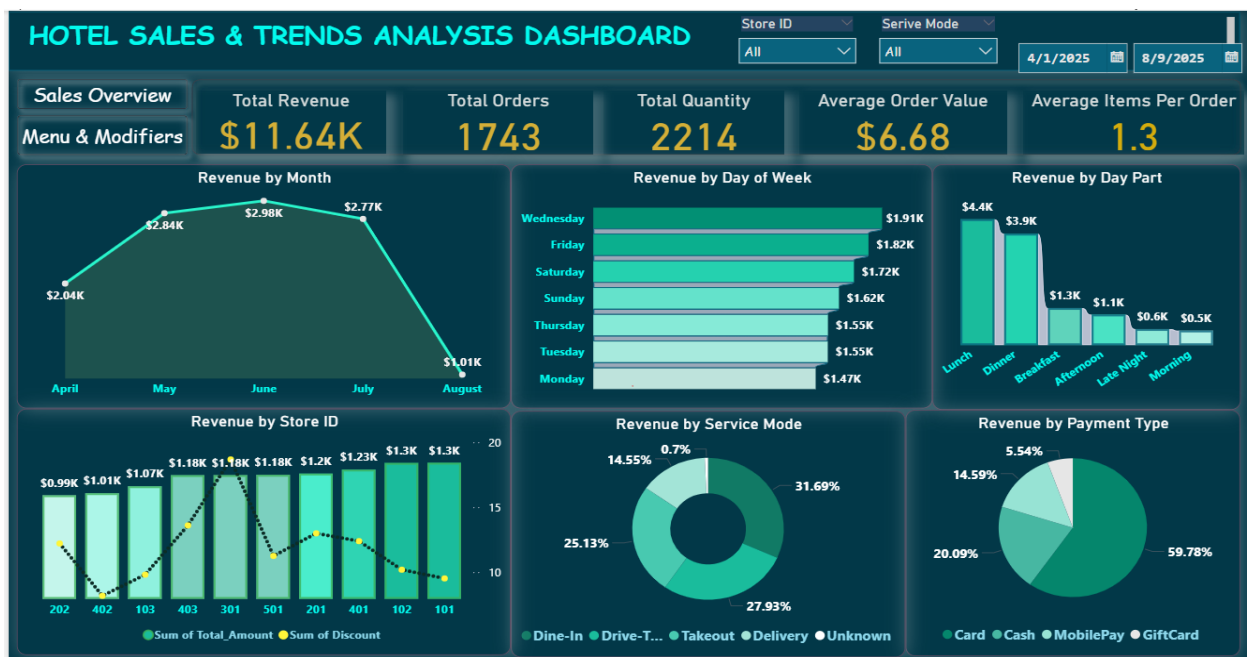
- Displays revenue distribution by payment method.
- Helps understand customer payment preferences.



Overall Business Insight (For page 1)

This page provides a consolidated view of overall sales performance. It helps management monitor key KPIs, compare store performance, and evaluate service mode contribution. The dashboard supports quick and data-driven decision-making.

Sales Overview Page



Page 2 – Customer Behavior & Trends Analysis

Objective

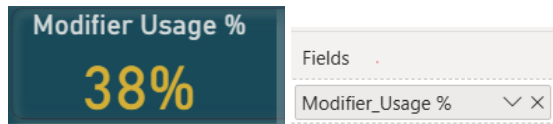
The objective of this page is to analyze customer purchasing behavior and understand how menu items and modifiers influence sales performance.

This page focuses on identifying customer customization patterns and product preferences.

KPI Cards & Purpose

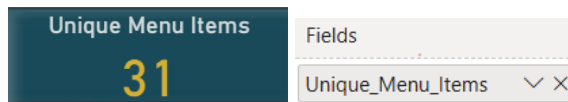
1.Modifier Usage %

- Shows percentage of orders that include at least one modifier.
- Helps measure customization behavior.



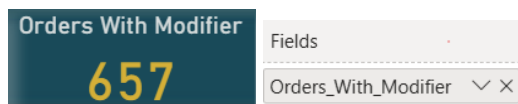
2. Unique Menu Items

- Displays number of distinct items sold.
- Indicates product variety and coverage.



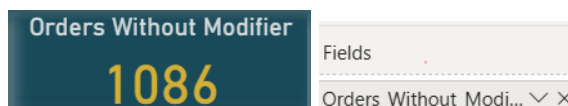
3.Orders With Modifier

- Total orders where customers customized items.
- Helps evaluate upselling performance.



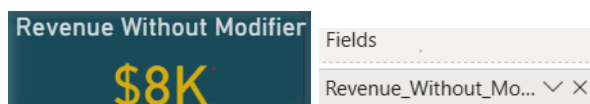
4.Orders Without Modifier

- Orders placed without customization.
- Helps compare standard vs customized demand.



5.Revenue Without Modifier

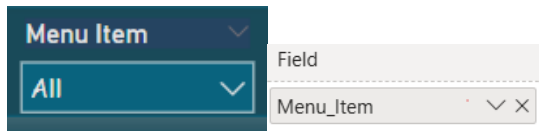
- Revenue generated from standard orders.
- Helps assess base product performance.



Slicer Used & Purpose

1.Menu Item Slicer

- Allows filtering of analysis based on a specific menu item.
- Helps analyze modifier usage and order trends for individual products.
- Enables deeper understanding of product-level customization behavior.



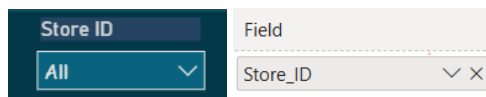
2.Date Slicer

- Allows filtering of data by specific date range.
- Helps analyze performance over different time periods.



3.Store ID Slicer

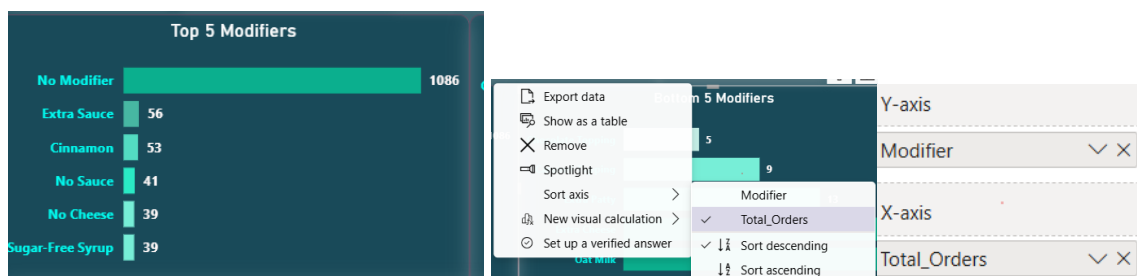
- Filters sales data by store.
- Helps compare performance across multiple stores.



Visual Analysis

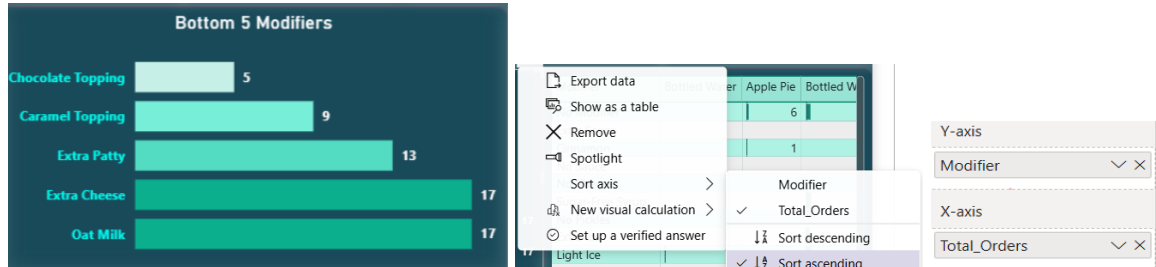
1.Top 5 Modifiers

- Identifies most frequently used add-ons.
- Helps understand popular customer preferences.



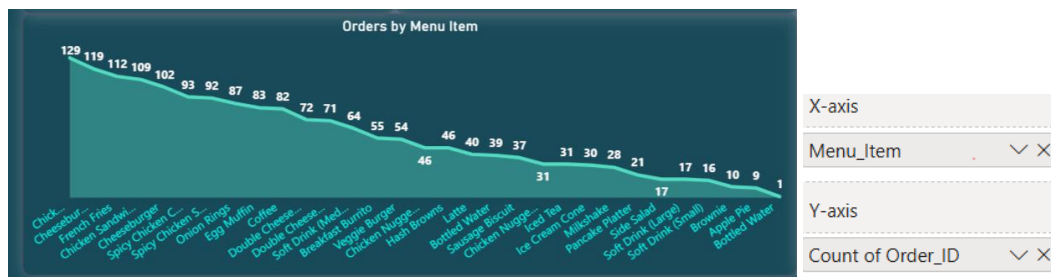
2.Bottom 5 Modifiers

- Shows least preferred modifiers.
- Useful for menu optimization decisions.



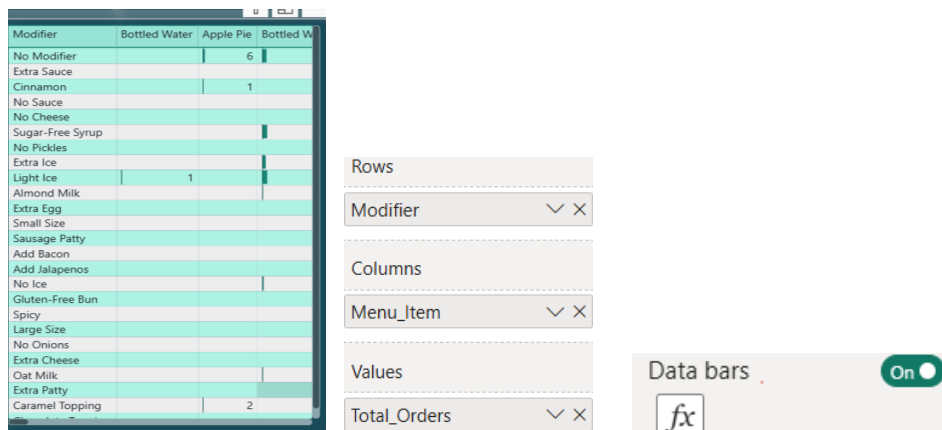
3.Orders by Menu Item

- Displays demand ranking of menu items.
- Helps identify top-selling and low-performing products.



4.Modifier Breakdown Table

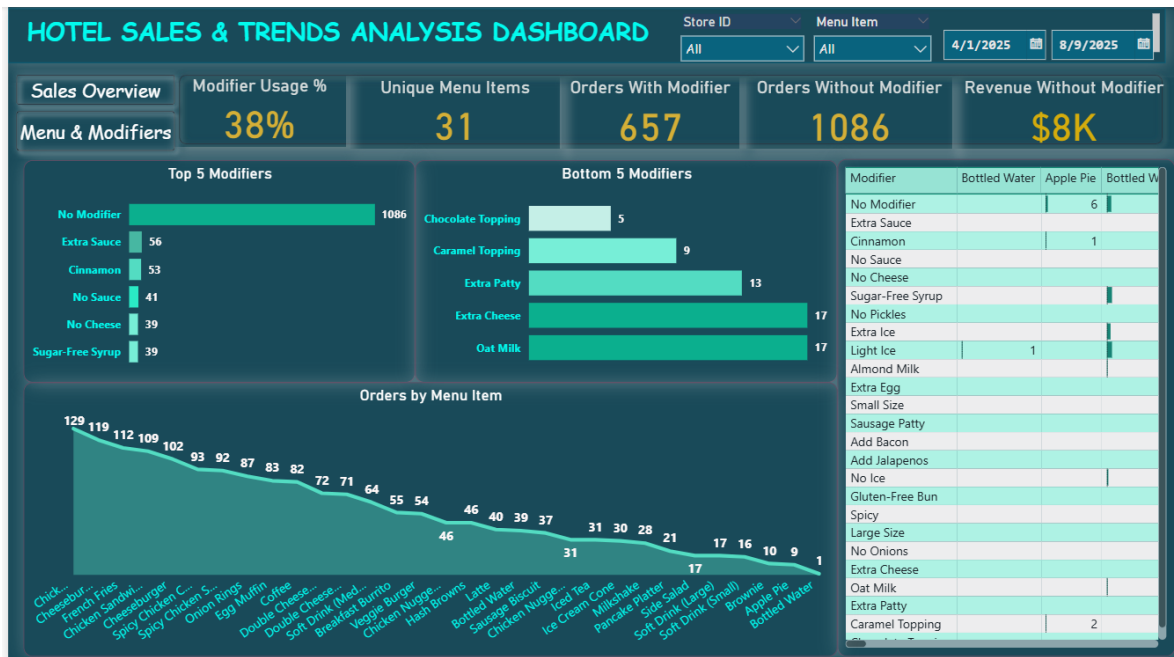
- Provides detailed relationship between menu items and modifiers.
- Supports deeper analytical exploration.
- Feature Applied:** Data Bars (Conditional Formatting)



Overall Business Insight (For Page 2)

This page helps understand customer customization behavior and menu preferences. It enables management to optimize menu offerings, improve upselling strategies, and eliminate low-performing modifiers.

Customer Behavior & Trends Analysis Page

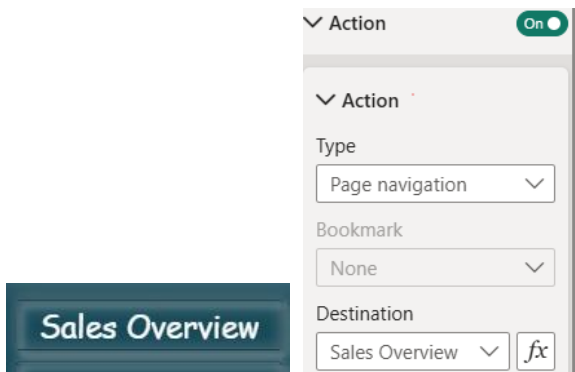


Page Navigation & Interactive Design

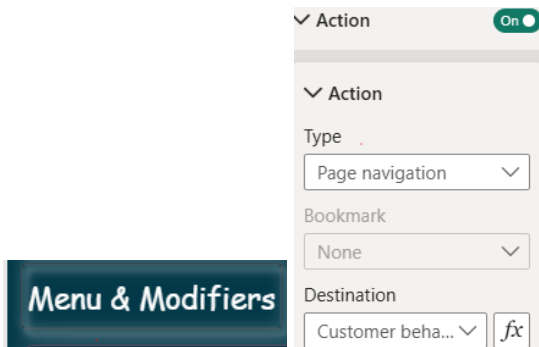
Improve user experience, interactive navigation buttons were created on each page.

Two buttons are provided:

- Sales Overview



- Menu & Modifier Analysis



These buttons allow users to seamlessly navigate between the two dashboard pages.

Key Insights Identified

1. Revenue peaked during June, indicating strong mid-year performance.
2. Lunch and Dinner contributed the highest revenue among all day parts.
3. Certain stores generated higher revenue but also offered higher discounts, showing promotional impact.
4. A significant portion of orders were placed without modifiers, indicating preference for standard menu items.
5. Only 38% of orders included customization, suggesting moderate modifier usage.
6. Some modifiers had very low demand, indicating potential menu optimization opportunity.

Project Conclusion

This project successfully transformed raw hotel transaction data into an interactive and insight-driven Power BI dashboard.

Through structured data cleaning in Excel, DAX measure creation, and analytical visualization design, the dashboard provides a clear understanding of sales performance and customer behavior.

The analysis identified revenue trends, store-level discount impact, service mode contribution, and modifier usage patterns. These insights support data-driven decision-making in pricing strategy, promotional planning, and menu optimization.

Overall, this project demonstrates practical skills in data preparation, analytical thinking, and business-focused dashboard development.

Final Dashboard View

